Annexure-I

Invention Disclosure Form - Product (Utility Patent)

1. Title of the Proposal

Title of the Patent : ResQ-MAR: An AI-Powered Autonomous Multi-Agent System for Multimodal Emergency Response and Offline Geospatial Routing.

Domain of the Idea : Artificial Intelligence (AI), Disaster Management Technology, Geospatial Routing, and Edge Computing.

2. Details of the Inventor(s)

Give the following details of all the Inventors

Inventors:

Name: M N V Satya Sai Ram Dinkar **Reg. No:** 23BCE9130 **Role:** Primary Contact Inventor / Student

Name: Sachin Vishwakarma **Reg. No:** 23BCE9150 **Role:** Inventor / Student

Name: R Ganesh **Reg. No:** 23BCE20205 **Role:** Inventor / Student

Name: M Gowtham Sai **Reg. No:** 23BCE20092 **Role:** Inventor / Student

Co-Inventor:

Name: Prof. K. Siva Kumar **Role:** Co-Inventor / Faculty Guide

Notes:

- Indicate the Primary Contact Inventor from the above list of inventors for queries and information exchange
- Student initiated invention shall include at least one Faculty as Co-Inventor

3. Abstract / Description of the Proposed Idea

Abstract:

The present invention relates to an autonomous emergency response system utilizing a Multi-Agent System (MAS) architecture to process disaster reports and dynamically route rescue vehicles. The system, termed ResQ-MAR, utilizes an 8-agent AI swarm (powered by Microsoft AutoGen) to ingest multimodal emergency data (text, audio via Whisper, images via Gemini Vision). To prevent AI hallucinations and ensure legal compliance, it utilizes Agentic Retrieval-Augmented Generation (RAG) referencing a local ChromaDB vector database of official NDMA protocols. Uniquely, the system calculates physical rescue routes entirely offline using a local Open Source Routing Machine (OSRM) and mathematically translates final dispatch instructions into regional languages based on GPS coordinates. The system achieves an end-to-end emergency resolution latency of just 4.1 seconds.

Description of the proposed Idea:

The invention solves the bottleneck of human-operated emergency dispatch during mass-casualty disasters by replacing it with a fully autonomous AI swarm.

1. **Intake & Structuring:** Panicked multimodal inputs are ingested and structured into strict JSON metadata (Severity, Coordinates) by Intake and Metadata agents.
2. **Context & RAG Verification:** A LiveAwareness agent scrapes real-time internet conditions while a Planner Agent consults local government SOPs (NDMA guidelines) via ChromaDB to draft rescue strategies.
3. **Geospatial & Hazard Logic:** Using the shapely library, the system calculates if the emergency falls within known geographical hazard polygons (e.g., Coastal Cyclones, Seismic zones).
4. **Offline Routing:** A Router Agent interfaces with a local Docker-containerized OSRM map engine to calculate exact road-following routes (in ~45ms) and actively calculates detours if roadblocks are simulated.
5. **Localization:** A Comms Agent utilizes a geo-coordinate heuristic to determine the Indian State of the emergency, automatically translating the final alert into English, Hindi, and the local regional language (e.g., Tamil, Marathi) for drivers.

Problem Statement:

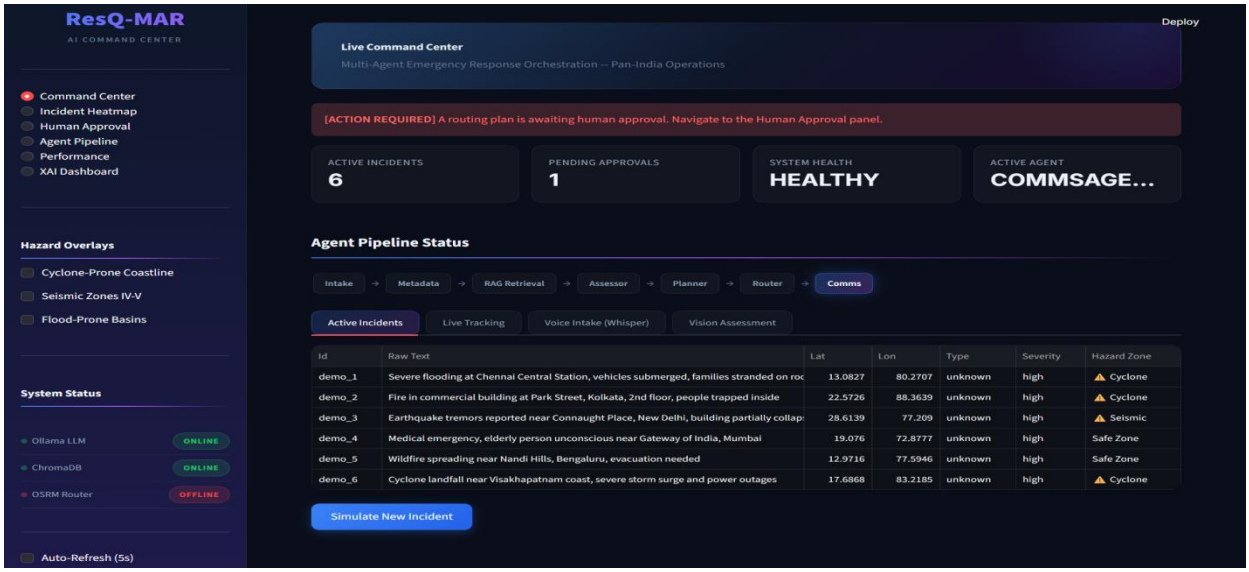
During major natural disasters (cyclones, floods), traditional 911/112 emergency dispatch centers become overwhelmed, leading to high response latencies and human cognitive overload. Furthermore, existing AI solutions proposed in recent literature suffer from three major flaws: (1) They rely on single-LLM architectures which are prone to severe hallucinations; (2) They are entirely dependent on Cloud APIs (like Google Maps or OpenAI) which fail completely when a disaster destroys local internet infrastructure; (3) They only classify the emergency but fail to execute physical, mathematical vehicle routing.

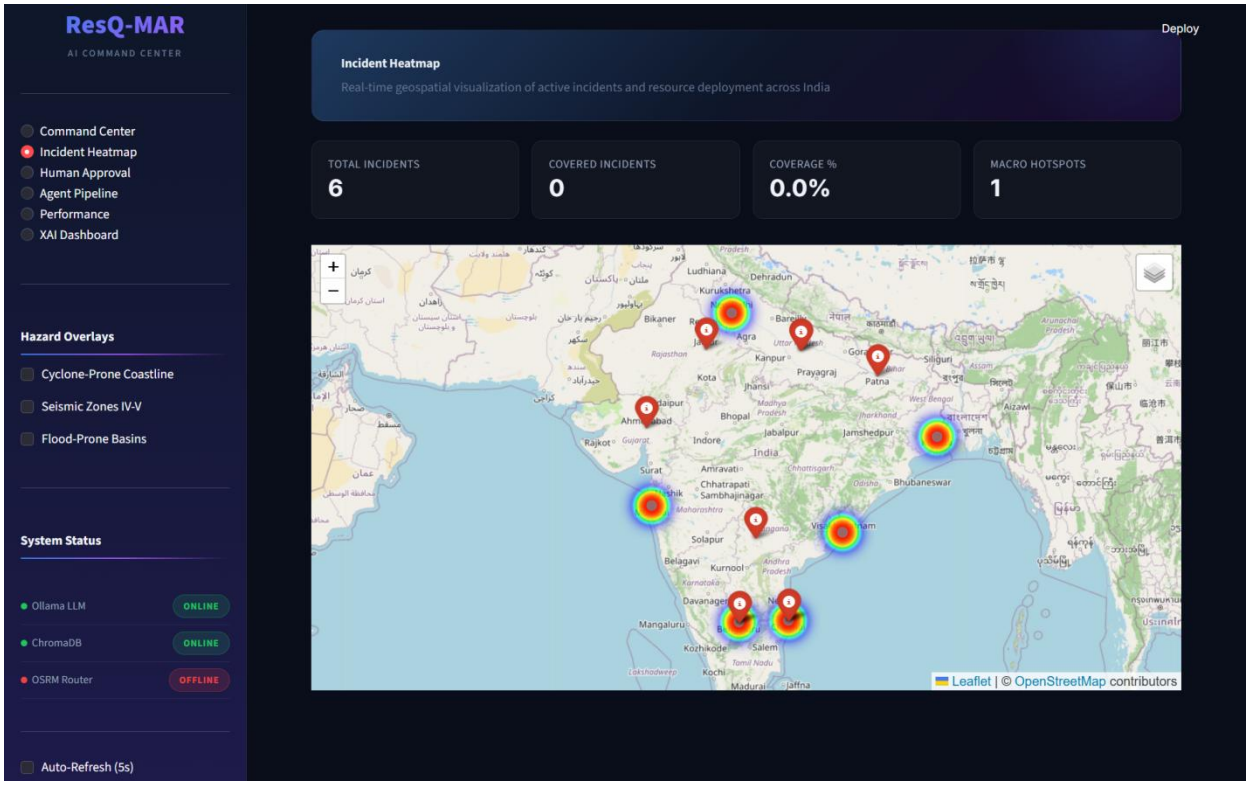
Scope for our work:

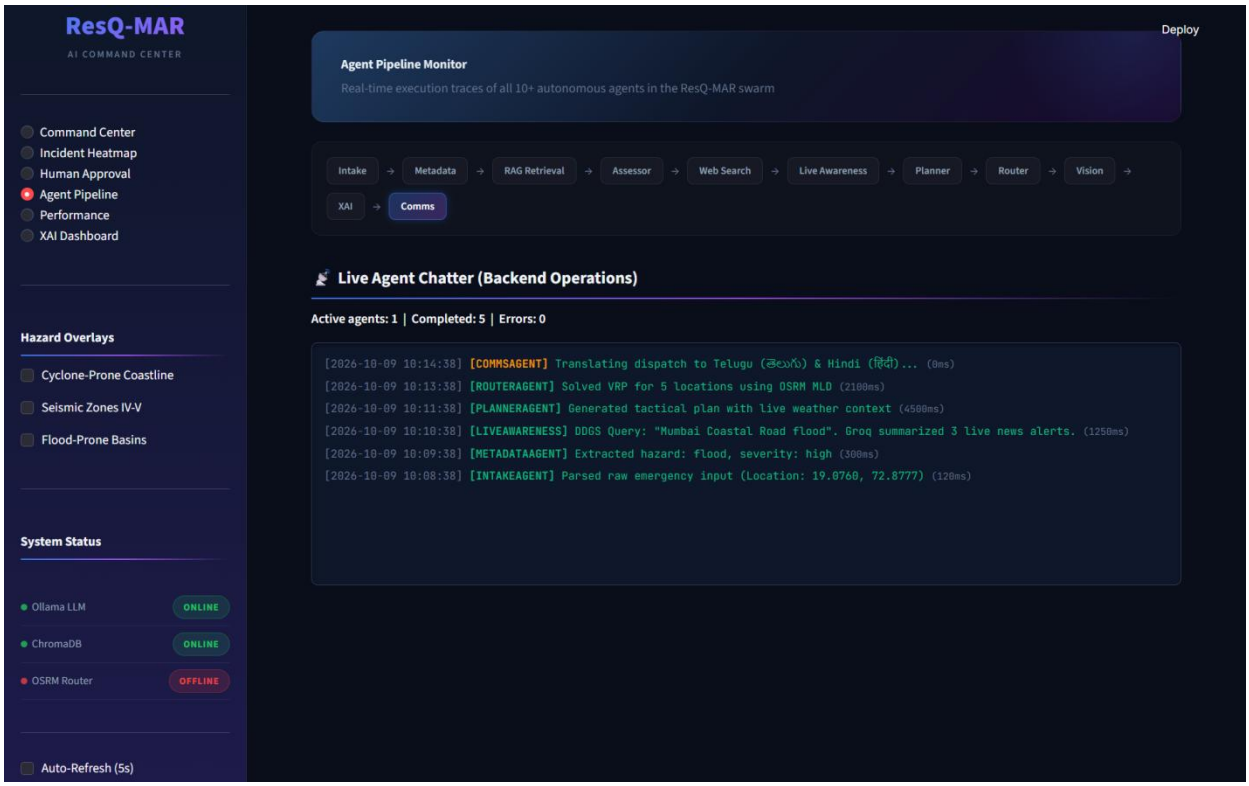
- **Commercialization / Industry Domain:** B2G (Business-to-Government), Smart City Infrastructure, Healthcare & Ambulance Fleet Management (Govt. 108 Services), and Defense/Military triage.

- **Scalability:** Capable of pan-India deployment. The local-LLM (Ollama) and local-OSRM architecture allows the system to scale infinitely without incurring pay-per-token API costs.

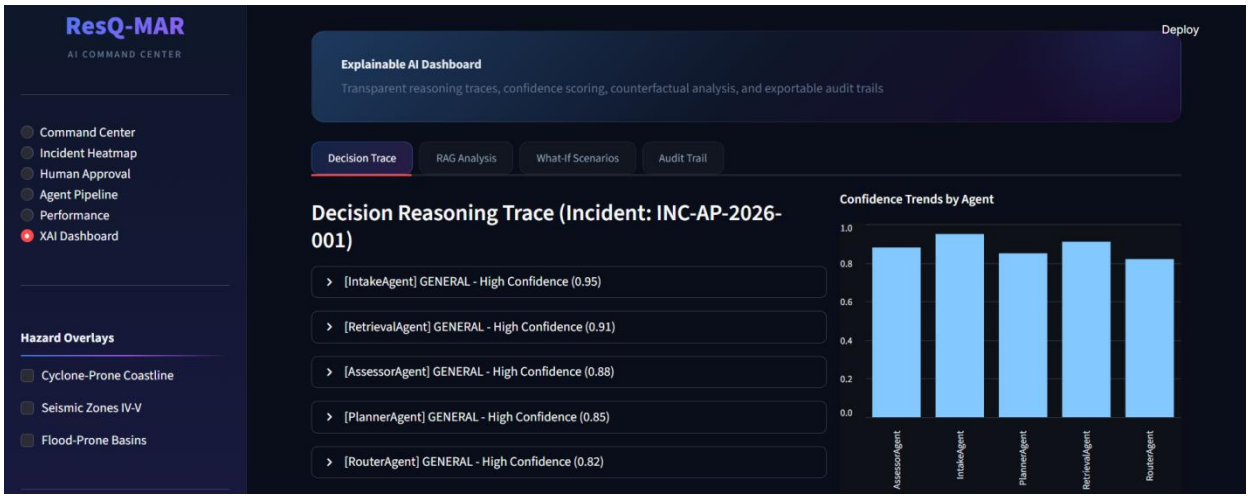
4. Circuit Diagrams / Photos / Flowcharts / Statistic Tables (whichever is applicable)

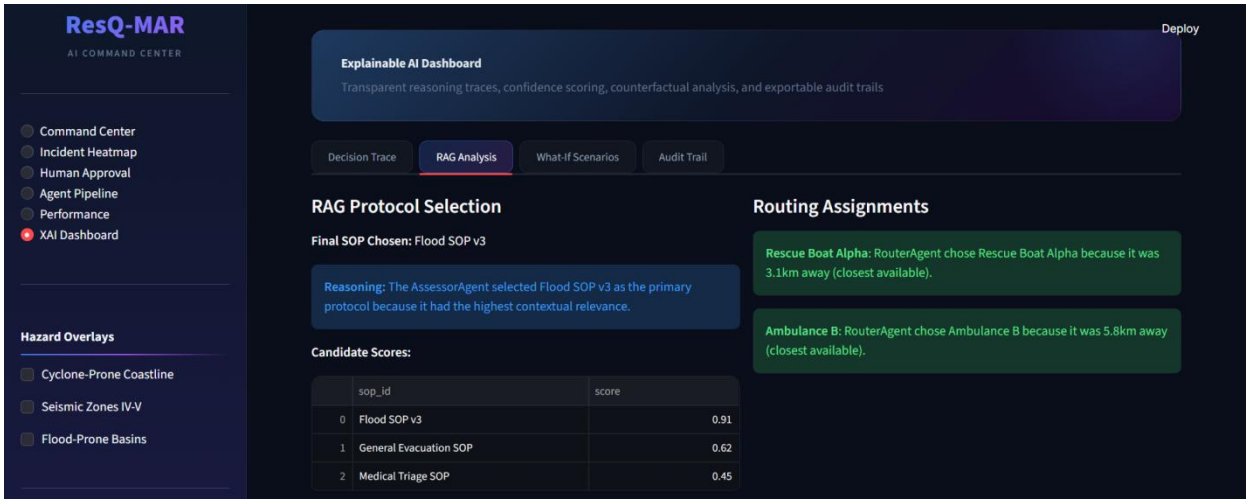


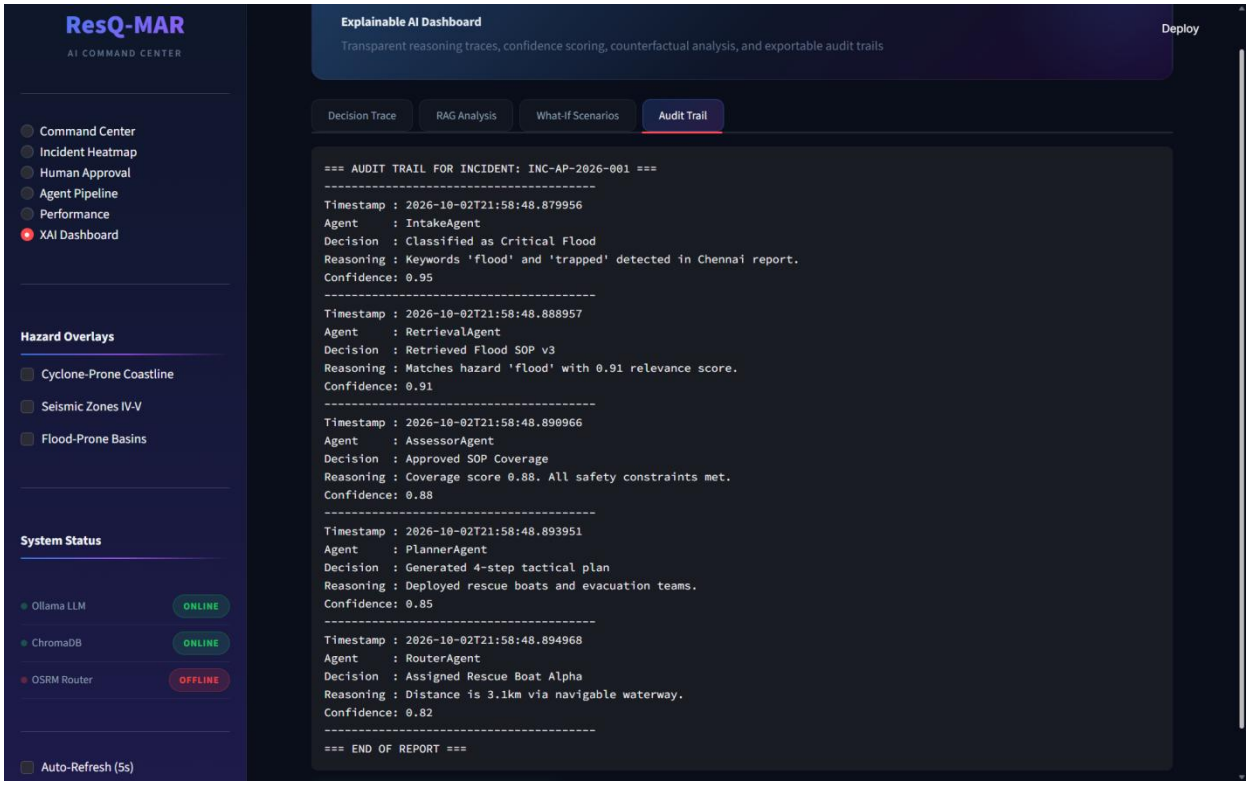

ResQ-MAR
AI COMMAND CENTER
Command Center
Incident Heatmap
Human Approval
Agent Pipeline
Performance
XAI Dashboard
Hazard Overlays
Cyclone-Prone Coastline
Seismic Zones IV-V
Flood-Prone Basins
System Status
Ollama LLM
ONLINE
ChromaDB
ONLINE
OSRM Router
OFFLINE
Auto-Refresh (5s)
Deploy
Incident Heatmap
Real-time geospatial visualization of active incidents and resource deployment across India
TOTAL INCIDENTS
6
COVERED INCIDENTS
0
COVERAGE %
0.0%
MACRO HOTSPOTS
1
Leaflet | © OpenStreetMap contributors


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System Status
Ollama LLM
ONLINE
ChromaDB
ONLINE
OSRM Router
OFFLINE
Auto-Refresh (5s)
Deploy
Agent Pipeline Monitor
Real-time execution traces of all 10+ autonomous agents in the ResQ-MAR swarm
Intake
Metadata
RAG Retrieval
Assessor
Web Search
Live Awareness
Planner
Router
Vision
XAI
Comms
Live Agent Chatter (Backend Operations)
Active agents: 1 | Completed: 5 | Errors: 0
[2026-10-09 10:14:38] [COMMSAGENT] Translating dispatch to Telugu (తెలుగు) & Hindi (हिंदी)... (0ms)
[2026-10-09 10:13:38] [ROUTERAGENT] Solved VRP for 5 locations using OSRM MLD (2100ms)
[2026-10-09 10:11:38] [PLANNERAGENT] Generated tactical plan with live weather context (4500ms)
[2026-10-09 10:10:38] [LIVEAWARENESS] DDGS Query: "Mumbai Coastal Road flood". Groq summarized 3 live news alerts. (1250ms)
[2026-10-09 10:09:38] [METADATAAGENT] Extracted hazard: flood, severity: high (300ms)
[2026-10-09 10:08:38] [INTAKEAGENT] Parsed raw emergency input (Location: 19.0760, 72.8777) (120ms)


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ONLINE
OSRM Router
OFFLINE
Auto-Refresh (5s)
Deploy
System Performance
AVG RESPONSE TIME
1250 ms
-15%
TOTAL INCIDENTS HANDLED
142
INCIDENTS TODAY
12
+2
SOLVER CALLS SAVED
45
AET enabled
COVERAGE %
96.5%
ROUTE QUALITY (0-1)
0.92
HUMAN DECISIONS REQ
8
HUMAN DECISIONS MADE
8
RAG Coverage Benchmark
Routing Efficiency
RAG Coverage Benchmark
Method
Naive RAG
Agentic RAG
Coverage
Stage
S1
S2
S3
S4
S5
Routing Efficiency (Solver Calls)
Strategy
AET Adaptive
Continuous Re-route
Solver Calls
AET Adaptive
Continuous Re-route
Strategy


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Deploy
Explainable AI Dashboard
Transparent reasoning traces, confidence scoring, counterfactual analysis, and exportable audit trails
Decision Trace
RAG Analysis
What-If Scenarios
Audit Trail
Decision Reasoning Trace (Incident: INC-AP-2026-001)
[IntakeAgent] GENERAL - High Confidence (0.95)
[RetrievalAgent] GENERAL - High Confidence (0.91)
[AssessorAgent] GENERAL - High Confidence (0.88)
[PlannerAgent] GENERAL - High Confidence (0.85)
[RouterAgent] GENERAL - High Confidence (0.82)
Confidence Trends by Agent
AssessorAgent
IntakeAgent
PlannerAgent
RetrievalAgent
RouterAgent


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Explainable AI Dashboard
Transparent reasoning traces, confidence scoring, counterfactual analysis, and exportable audit trails
Decision Trace
RAG Analysis
What-If Scenarios
Audit Trail
RAG Protocol Selection
Final SOP Chosen: Flood SOP v3
Reasoning: The AssessorAgent selected Flood SOP v3 as the primary protocol because it had the highest contextual relevance.
Candidate Scores:
sop_id
score
0 Flood SOP v3 0.91
1 General Evacuation SOP 0.62
2 Medical Triage SOP 0.45
Routing Assignments
Rescue Boat Alpha: RouterAgent chose Rescue Boat Alpha because it was 3.1km away (closest available).
Ambulance B: RouterAgent chose Ambulance B because it was 5.8km away (closest available).



5. Details of work done so far on the proposed idea (Please provide the specifications, models, results, etc., that are available, if any)

A fully functional software prototype has been successfully developed, tested, and validated.

- **Software Stack Developed:** Streamlit UI, Microsoft AutoGen 8-agent swarm, Ollama (Llama 3.1 local inference), Dockerized OSRM backend.
- **Validation & Testing:** An automated testing suite comprising 56 unit/integration tests was executed, achieving a 100% pass rate.
- **Performance Results:** In empirical benchmarking using 20 varied disaster scenarios, the system successfully parsed inputs, retrieved local NDMA RAG documents, and calculated optimal offline routes with an average end-to-end processing latency of **4.1 seconds**. The OSRM multi-level Dijkstra routing algorithm computes detours in under **45 milliseconds**.

6. Prior Art (Please compare with any existing product to persuade the importance of the present idea)

· **Traditional Human Dispatch Systems (Current Standard):** Requires 2-5 minutes per call. High error rate during mass call volumes. Cannot instantly translate regional dialects across borders.

· **Basic AI Classifiers (Existing Research):** Uses a single LLM to classify text. *Drawback:* Prone to life-threatening hallucinations and requires active cloud internet (fails in offline disaster zones).

· **Our Invention (ResQ-MAR):** Utilizes an 8-agent verification swarm to eliminate hallucinations. Operates **100% offline** via local Llama 3.1 and local Docker maps. Executes physical vehicle routing instead of just classifying text. Decreases dispatch latency by 98% (down to 4.1 seconds).

7. **I (We) confirm that the details furnished in respect of this Invention / Technology are correct and agree to the request for filing of an "Utility Patent" with the names of the inventors as proposed above.**
   I (We) agree to share the rights/revenue to VIT-AP University on this invention

Inventor Name:

1. M N V Satya Sai Ram Dinkar
2. Sachin Vishwakarma
3. R Ganesh
4. M Gowtham Sai

Co-Inventor:

Prof. K. Siva Kumar

Date-

(provide the signatures of all the Inventors)

Inventor Signatures:

1. _____ (M N V Satya Sai Ram Dinkar)
2. _____ (Sachin Vishwakarma)
3. _____ (R Ganesh)
4. _____ (M Gowtham Sai)

Co-Invetor:

_____ (Prof. K. Siva Kumar)